

1:20 Scale Self Parking Vehicle

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ABSTRACT

Current autonomous parking technology is expensive due to the types of sensors commonly used. As a proof of concept for a cheaper system, this project develops an autonomous parking system using ultrasonic sensors. The goal of this project is to build a representative 1:20 scale car and autonomously perpendicularly reverse park into a parking space purely using ultrasonic sensors for object detection. If this technology can be scaled to full-size vehicles, self-parking technology can be more widely accessible for everyone.

METHODOLOGY

The vehicle is a 1:20 scale model of a real car; specifically a Tesla Model Y. The car features front Ackerman steering geometry made out of 3D printed parts and powered by an MG90S servo. The rear wheels are powered by an N20 DC motor using a TB6612FNG motor driver and a 3D printed differential. For object detection, four RCWL-1601 ultrasonic sensors are mounted around the vehicle. All of these components are controlled by an ESP32S3 and powered by a 5V 1A portable power bank.

While driving forward, the vehicle drives at a max speed slightly above 180 mm/s—the equivalent of 8 mph when scaled to a full-size vehicle. While parking, the motor is limited to around 100 RPM, which is roughly 65 mm/s—the equivalent of 3 mph when scaled to a full-size vehicle.

RESULTS

The finalized design functions as intended; however, there are some limitations to this technology when compared to other similar technologies:

- Cars must be parked directly on both sides of the target parking spot to ensure the vehicle is centered.
- A car must be directly behind the parking space to ensure the vehicle does not reverse too far back.

Despite these limitations, this technology still allows for autonomous parking at a cheaper cost than these other technologies. Additionally, these can be mitigated with further development of this technology.

MECHANICAL DESIGN

The steering setup is made up of six linkages. The servo driving the center linkage is allowed to rotate up to 18° left and right, allowing for the inner wheel to rotate up to 35°; the same as many real world vehicles.

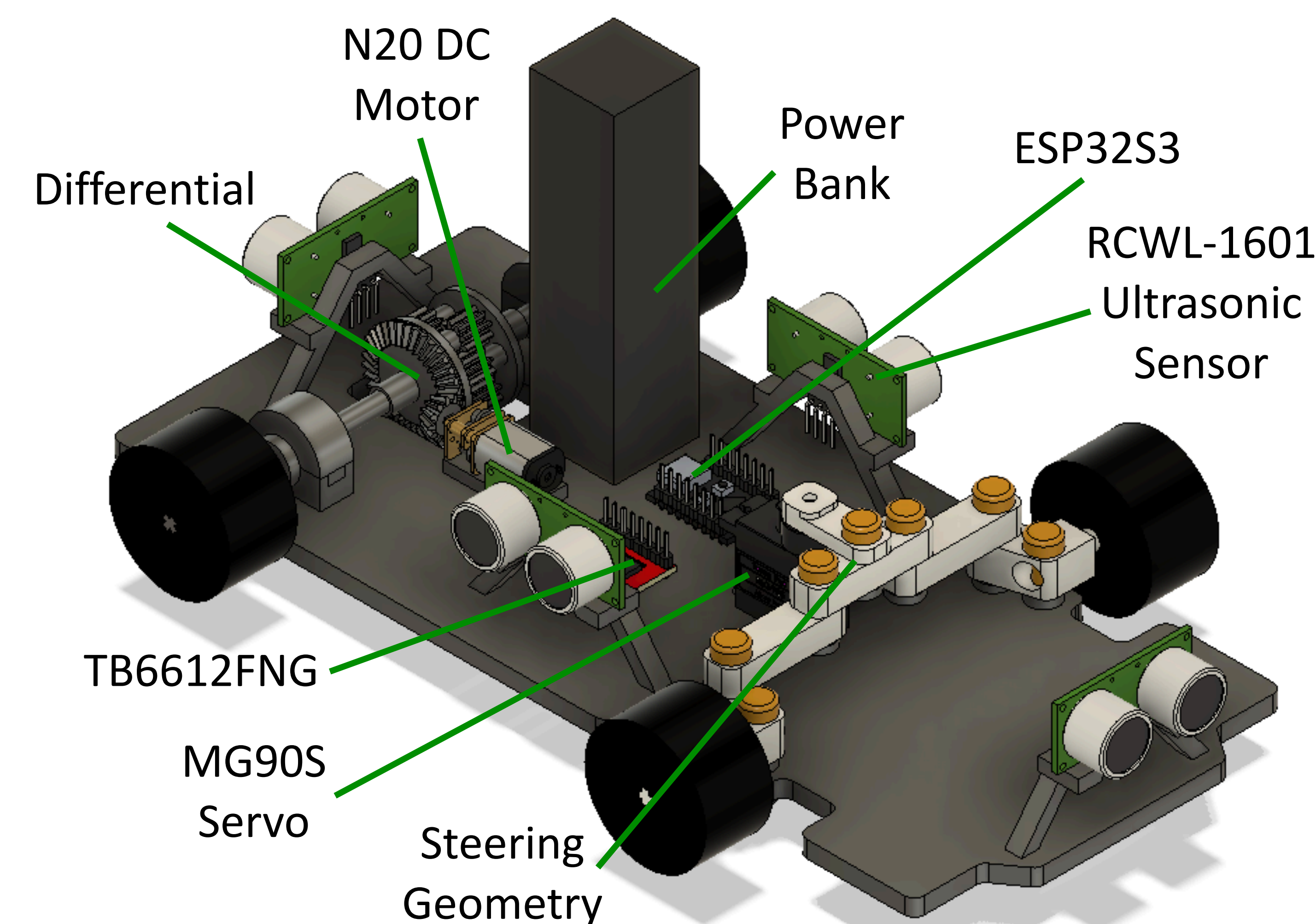
The differential design used is a spur gear differential. Each axle has a spur gear on its end, called a sun gear. The sun gears have sets of smaller planet spur gears around them, connecting the two sides while also allowing them to freely spin. This design uses three sets of planet gears in order to distribute the load. A large bevel gear is also mounted on the side of the housing for power. All of these parts were 3D printed out of PETG, with 6 walls and 50% infill. A tolerance of 0.25mm is used on all of the gears to ensure that they can rotate without too much friction.

AUTONOMY INTEGRATION

The vehicle has four ultrasonic sensors. The front sensor ensures the vehicle does not collide with objects in front of it. The side sensors detect parking spaces and also help avoid collisions while turning into a parking space. The rear sensor stops the vehicle from colliding with objects at the end of the parking space.

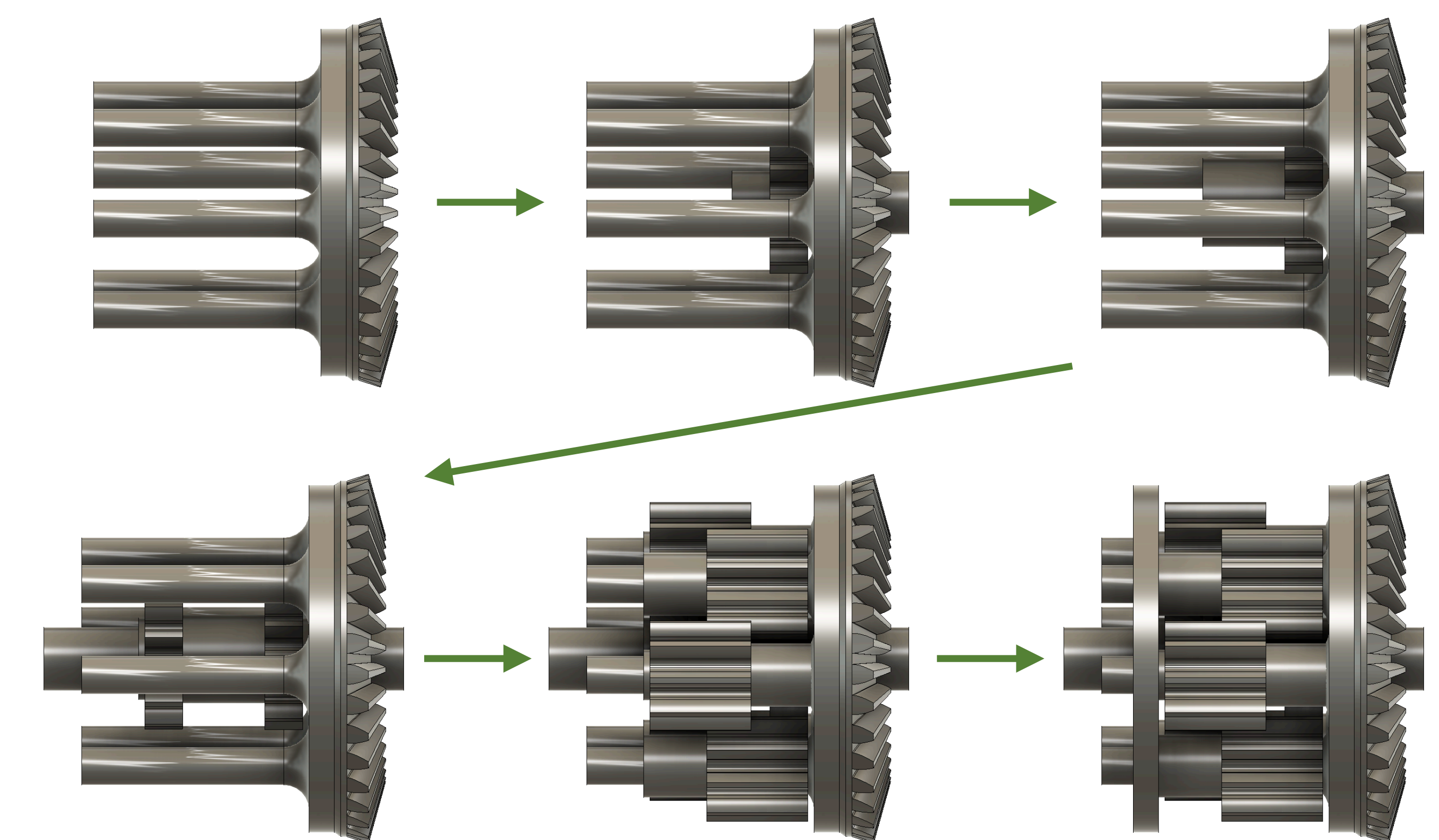
All components are controlled by an ESP32S3. The vehicle starts in a “searching” mode, driving forward until the sensors detect an open spot, while using the front sensor to avoid head-on collisions. Once an open spot is detected, the “parking” mode is initiated, turning the servo and slowing the speed, to position the car for reversing. The servo and motor then reverse directions as the vehicle reverses into the spot, while the sensors prevent any collisions.

FINALIZED CAD MODEL



DIFFERENTIAL CAD MODEL

Below are several images of the finalized CAD model for the differential, showing how the part is assembled.



STEERING GEOMETRY CAD MODEL

Below is the finalized CAD model for the steering geometry, showing the wheels fully turned left, centered, and fully turned right, respectively.

